INCH-POUND
MIL-PRF-49175C(CR)
9 October 1997
SUPERSEDING
MIL-C-49175B(CR)
20 February 1992

PERFORMANCE SPECIFICATION

COOLER, CRYOGENIC, MECHANICAL HD-1033B/UA AND HD-1033C/UA

This specification is approved for use within the Communications-Electronics Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of common module with 2 interchangeable models; Cooler, Cryogenic, Mechanical, HD-1033B/UA and HD-1033C/UA

2. APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army, Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703-5023 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The cooler, Cryogenic, Mechanical HD-1033B/UA and HD-1033C/UA, referred to herein as the cooler, is a closed-cycle refrigerator for cooling the infrared detector and interfacing with the Dewar of the infrared detector-Dewar package in an infrared system.

3.2 Interface requirement. The cooler, Cryogenic Mechanical HD-1033B/UA shall conform to Drawing SM-D-969477 and SM-D-971500 for interface requirements.

3.2.1 Weight. The weight of the cooler shall not exceed 3.8 pounds.

3.3 First article. The contractor shall furnish first article assemblies in accordance with 4.2.

3.4 Materials, parts, and processes. The contractor shall select the parts, materials and processes but these materials shall be capable of meeting all of the operational and environmental performance requirements specified herein.

3.5 Performance characteristics. The cooler shall meet the performance characteristics specified herein when adequate heat sinking or convective cooling is provided to ensure that any point on the cooler cylinder head shall fall within 5°C to 15°C above ambient air temperature.

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3.5.1 Cooling capacity. The cooler with thermal load (copper mass) shall provide the minimum refrigeration capacity at 80° K maximum as shown in figure 1, curve A.

3.5.2 Cooldown time. The cooldown time to reach a cold tip temperature of 80° K from ambient with a 1440, (+25, -0) joule copper thermal mass (from 300° K to 80° K) shall be in accordance with figure 2.

3.5.3 Input power. The total input power to the cooler shall not exceed that shown in figure 3 with 1440, (+25, -0) joule thermal mass after cold tip temperature has stabilized with the heat load shown in figure 1, curve A. The input voltage shall be 117, ±2 volts alternating current (VAC), 400, ±20 hertz (Hz) power source.

3.5.4 Acoustic noise. The cooler noise emission shall not be more than the values listed below, when measured at a distance of 3 feet.

Sound pressure values

Center Frequency (Hz)	Octave band (Hz)	Maximum sound pressure level(dB), ref.0.0002 microbar
125	87-175	45
250	175-350	45
500	350-700	48
1000	700-1400	55
2000	1400-2800	60
4000	2800-5600	65
8000	5600-11200	65

3.5.5 Leak rate. The leak rate of the cooler shall not be greater than 2.5×10^{-7} cubic centimeters per second (cc/sec) helium equivalent at an ambient temperature of +23°C, ±5°C.

3.5.6 Vibration output. The maximum amplitude, at the ambient temperature of +23°C ±5°C, of the vibrational force induced by the cooler at a frequency of 26Hz ±4Hz or any of the next eight harmonics along the compressor piston and coldfinger axes shall not exceed 0.35 pound (semi-amplitude) and shall not exceed 0.5 pound (semi-amplitude) along the motor axis. The maximum amplitude of the torque induced by the cooler about the motor, piston and coldfinger axes at 26Hz ±4Hz or any of the next eight harmonics shall not exceed:

a. 4.0 In-oz (semi-amplitude) compressor piston axis
b. 4.0 in-oz (semi-amplitude) coldfinger axis
c. 15.0 in-oz (semi-amplitude) motor axis.

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3.5.7 Restart. The cooler shall restart after having cooled to the operating temperature of at least 80°K at any ambient from -54°C to +71°C.

3.6 Environmental conditions.

3.6.1 Temperature shock. The cooler shall not be damaged (see 6.3.1) by sudden changes in temperature between -62°C and +95°C.

3.6.2 High temperature. The cooler shall not be damaged by storage to +95°C or operation up to +71°C.

3.6.3 Low temperature. The cooler shall not be damaged by operation to -54°C or storage to -62°C.

3.6.4 Shock. The cooler shall not be damaged by high intensity shocks of 100 g's peak amplitude with 11 milliseconds duration and by bench handling tests.

3.6.5 Vibration. The cooler shall not be damaged by vibration over a frequency spectrum at the specified g levels and amplitudes shown in Figure 4.

3.7 Burn-in. Each cooler delivered shall have accumulated a minimum of 12 hours burn-in over the temperature profile in accordance with figure 5.

3.8 Reliability. The cooler shall have a lower mean-time-to failure (MTTF) of at least 1,000 hours.

3.9 Nameplates and product marking. The cooler shall have a nameplate marked with the National Stock Number and the manufacturer's part number.

3.10 Workmanship. Coolers shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect their life, serviceability, interchangeability, or appearance.

3.11 Labeling and marking. All labeling and marking shall be clear and legible throughout all the tests specified herein. Labeling and marking shall be black. Jackets may have the labeling and marking engraved, or die stamped, in which case it may be the same color as background. The top surface of the cooler shall be marked with the following in 0.12 inch Gothic characters:

80063 Assy-
HD-1033C/UA or HD-1033B/UA (as appropriate)

4. VERIFICATION

4.1 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.2.1)
b. Conformance inspection (see 4.3).

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4.2 Inspection conditions. Except as otherwise specified herein, all examinations and tests shall be performed at a temperature of 80 ± 20°F.

4.2.1. First article inspections. First article inspections shall be performed by the contractor after award of contract and prior to production. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. Inspections required for First Article are listed in Table I.

4.2.2 Failure. If one or more sample coolers fail to meet any of the first article requirements and tests, the contractor shall immediately notify the government of the failure, the contractor shall determine the root cause of the failure and take appropriate corrective action. The contractor is required to submit a new set of first article samples which have incorporated the charges for inspection. A description of the failures and corrective action taken should be included in the first article inspection reports.

4.2.3 Test. Upon successful completion of the inspections specified in 4.2.1, four first article coolers shall be subjected to reliability testing and 2 first article coolers shall be subjected to all other tests listed in Table I. Tests may be conducted in any order. Failure of any test shall constitute first article failure.

4.2.3 1 Disposition of first article samples. First article samples shall not be considered as part of the procurement quantities.

TABLE I. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Cooling capacity	3.5.1	4.5.1
Cooldown time	3.5.2	4.5.2
Input power	3.5.3	4.5.3
Acoustic noise	3.5.4	4.5.4
Leak rate	3.5.5	4.5.5
Vibration output	3.5.6	4.5.6
Restart	3.5.7	4.5.7
Temperature shock	3.6.1	4.6.1
High temperature	3.6.2	4.6.2
Low temperature	3.6.3	4.6.3
Shock	3.6.4	4.6.4
Vibration	3.6.5	4.6.5
Reliability	3.8	4.8

4.3 Inspection procedures for verification. Unless otherwise specified herein, the cooler shall be operated at an ambient temperature of +23°C ±5°C. Adequate heat sinking or convective cooling shall be provided to ensure that any point on the cooler cylinder head shall fall within 5°C to 15°C above ambient air temperature. (See 3.6)

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4.4 Conformance inspection.

4.4.1 Inspection. Inspection shall be as specified in 4.2.1.

4.4.2 Tests.

4.4.2.1 Burn-in. Burn-in shall be in accordance with 4.7 and shall be performed upon each cooler that has passed the inspection of 4.5.1.

4.4.2.2 Group A inspection. Group A inspection shall be conducted on all coolers which have completed burn-in specified in specified in 4.4.2.1. Group A tests listed in table II may be performed in any order unless otherwise specified. Failure of any test shall be cause for rejection of that unit.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Cooling capacity	3.5.1	4.5.1
Cooldown time	3.5.2	4.5.2
Input power	3.5.3	4.5.3
Leak rate	3.5.5	4.5.5
Restart	3.5.7	4.5.7

4.4.2.3 Group B inspection. Group B inspections shall be conducted on coolers selected from units which have passed the tests in 4.4.2.2 (see 6.2h). The sample(s) shall be tested in accordance with the inspections listed in Table III. Samples shall be selected in accordance with 4.4.2.4.1. Group B tests listed in table IV may be performed in any order.

4.4.2.4.1 Sampling for group B inspection. Group B samples shall be selected as follows:

TABLE III. Group B inspection.

Inspection	Requirement Paragraph	Test Paragraph
Acoustic noise	3.5.4	4.5.4
Vibration output	3.5.6	4.5.6
Temperature shock	3.6.1	4.6.1
Low temperature	3.6.3	4.6.3
High temperature	3.6.2	4.6.2

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TABLE IV. Group B Sampling Plan.

Monthly Lot Size	Number of sample units monthly
1-99	1
100-299	3
300-499	5
500-999	7
1000 or more	10

4.4.2.4.2 Group B failures. Actions required relative to group B failures shall be as specified in the contract.

4.4.2.4.3 Disposition of group B samples. Group B samples shall be accepted on contract subsequent to the tests of table III.

4.4.2.5 Group C inspection. Group C inspection shall be conducted on coolers selected from units which have passed the tests in 4.4.2.2. Reliability inspection shall be in accordance with 4.8 and Table V.

TABLE V. Group C Inspection.

Inspection	Requirement Paragraph	Test Paragraph
Reliability	3.8	4.8

4.5 Test methods.

4.5.1 Cooling capacity. The cooling capacity shall be measured by attaching to the cooler a test Dewar and coldstation. The heat load shall be applied 30 minutes after achieving cooldown. Capacity shall be measured not less than 20 minutes later. The tests of paragraphs 4.5.1, 4.5.2 and 4.5.3 shall be performed using test equipment consisting of the following items to measure cooldown and refrigeration capacity:

1. Test dewar
2. Coldstation
3. Appropriate power supplies and meters
4. High vacuum source

    a. The test dewar shall be made of either brass or aluminum with a highly polished surface finish of at least an 8 RMS. The test dewar in conjunction with the high vacuum source shall enclose the coldfinger and coldstation in a vacuum of $1x10^{-5}$ torr or lower.

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b. The 1440 joule coldstation shall consist of the following elements as defined on drawing 54490 5004372: copper block, temperature sensor, resistor, thermally conductive epoxy and a clamp screw.

c. The coldstation assembly will have the following characteristics:

1. The temperature sensor and resistor will be bonded to the copper block with the thermally conductive epoxy.

2. The thermal energy which must be removed when cooling the coldstation from 300 K to 80 K shall be 1440, (+25, -0) joules. The mass of the coldstation shall be adjusted until it meets this requirement.

3. The resistor in the coldstation shall have a voltage applied to obtain a 1000, (+20, -0) mw thermal load. The applied voltage shall be such that the product of the voltage and the current of the resistor shall equal 1000,(+20,-0) mw.

4. The resistor and temperature sensor shall be connected to a vacuum feedthru by four teflon coated constantan wires. The wires shall be less than six inches in length and the resistance shall be less than one percent of the value of the resistor in the coldstation.

5. The coldstation is clamped to the coldfinger, sufficiently tight to effect good heat transfer without damaging the coldfinger.

6. Good thermal conductivity is assured by applying a metal oxide impregnated silicone thermal joint compound to the end of the coldfinger that is in contact with the coldstation.

7. A radiation shield shall not be used.

4.5.2 Cooldown time. Cooldown time shall be measured using the test Dewar and the copper mass described in 4.5.1. Failure to meet requirements of 3.5.2 shall constitute failure of this test.

4.5.3 Input power. Input power shall be measured when the cold tip temperature has stabilized below 80 K with heat load applied. Temperature stabilization has occurred when the output of the temperature sensor diode does not change more than .015 Vdc over a 5 minute period. Failure to meet requirements of 3.5.3 shall constitute failure of this test.

4.5.4 Acoustic noise. The cooler shall be set up for operation in an area where the background noise level is at least 5 dB below the sound level to be measured. The cooler shall be operated and sound pressure measurements made with the cooler oriented in 4 positions in the same plane with each position approximately 90 degrees apart. Measurements shall be made with an octave-band analyzer. Failure to meet requirements of 3.5.4 shall constitute failure of this test.

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4.5.5 Leak rate. The cooler shall be placed in a bell jar or suitable fixture connected to a helium mass spectrometer. The bell jar or fixture shall be evacuated to establish an inside-out test mode. Operating test shall not be performed during leak rate test. The leakage rate shall be measured at +23°C, ±5°C ambient. Failure to meet requirements of 3.5.5 shall constitute failure of this test.

4.5.6 Vibration output. The cooler shall be turned on for 15 minutes to allow it to reach equilibrium temperature. The cooler shall be vertically suspended such that the fundamental frequency of the cooler is much higher than the natural frequency of the suspension system. The static deflection of this arrangement shall be in excess of 6 inches and critical frequency (f_{cr}) of 1.3 Hz approximately. The acceleration along each axis shall be obtained by attaching an accelerometer to the center of the piston and motor faces and to the unit centered on the coldfinger axis. Moments around each axis shall be obtained by attaching two accelerometers (to the motor, piston face or coldfinger face) and coherently subtracting the two signals. This difference divided by the separation distance between the accelerometers yields the angular acceleration. Moment of inertia shall be obtained from the period of a bifilar pendulum containing the cooler. Peak forces are derived from the expression F=MA where F is the peak force, M is the cooler mass, and A is the measured acceleration at a given frequency. Peak torque is derived from the expression $T = I\wp$, where T is the peak torque, I is the moment of inertia about the principal axis, and $\wp$ is the measured angular acceleration for the applicable axis at a given frequency. Failure to meet requirements of 3.5.6 shall constitute failure of this test.

4.5.7 Restart. The cooler shall be operating at 80 K or less in a 23°C ± 5°C ambient. The power switch shall be turned off and the cooler motor shall be at rest. The power switch shall be turned on and the cooler must start within 5 seconds. This test shall be repeated a total of twenty times. The cooler shall also be tested at -54°C and -71°C using the above procedure during the high and low tests of 4.6. The restart test shall be performed after the cooldown and cooling capacity tests. Failure to meet the requirement of 3.5.7 shall constitute failure of this test.

4.6 Environmental tests. Unless otherwise specified, the environmental tests shall be performed in accordance with MIL-STD-810. The operating tests required before and after environmental tests shall be those specified in table II. During 4.6.2 and 4.6.3, operating test 4.5.1, 4.5.2, 4.5.3 and 4.5.7 shall be performed and must be successfully completed. Operating tests are not required during 4.6.1, 4.6.4 and 4.6.5 except that the unit under test shall have power applied during 4.6.4 and 4.6.5.

4.6.1 Temperature shock. The cooler shall be tested in accordance with MIL-STD-810, method 503, except that the temperature of step 2 shall be -54°C and the temperature of step 1 and step 4 shall be +95°C. Failure to meet the requirements of paragraph 3.6.1 shall constitute failure of this test.

4.6.2 High Temperature. The cooler shall be tested in accordance with MIL-STD-810, Method 501, Procedure I (see figure 6) except that the temperature of step 2 shall be +95°C. The temperature of step 4 and step 5 shall be +71°C. Failure to meet requirements of 3.6.2 shall constitute failure of this test.

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4.6.3 Low Temperature. The cooler shall be tested in accordance with MIL-STD-810, Method 502, Procedure I (see figure 7) except that the storage temperature of step 2 shall be -62°C and low operating temperature of step 4 shall be -54°C. Failure to meet requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Shock. The cooler shall be tested in accordance with MIL-STD-810, Method 516, Procedure IV, figure 516-1 (100 g's peak value at 11 milli-seconds time duration) and procedure 4. Failure to meet requirements of 3.6.4 shall constitute failure of this test.

4.6.5 Vibration. The cooler shall be tested in accordance with MIL-STD-810, Method 514, Procedure VIII except that the curve shown in figure 4 of this specification shall be used in lieu of the MIL-STD-810 curve. The sinusoidal cycle time shall be 120 minutes per axis. The dwell time shall be 1/6 of the cycling time at each resonance. Failure to meet requirements of 3.6.5 shall constitute failure of this test.

4.7 Burn-in. Each cooler shall be subjected to a minimum of 12 hours burn-in in accordance with figure 5. Failure of burn-in shall be defined as failure to pass the inspections listed in table II after burn-in. Any unit which fails and is repaired or is later altered, repaired or reworked, shall be resubjected to burn-in.

4.8 Reliability. A contractor prepared reliability test plan shall be used to demonstrate the reliability requirement of 3.8.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general and explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The cooler is intended for use in infrared systems.

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6.2 Acquisition Requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced.

c. Packaging requirements (see 5.1).

d. Group B and Group C inspection sample sizes should be established based on cumulative totals of concurrent contracts.

6.3 Definitions.

6.3.1 Damage. Breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection or component; and any degradation of cooler performance to values less than specified herein.

6.3.2 Lower mean time to failure. The term “Lower” is used to define that value which is the least operational capability and/or the maximum operating and support cost burden the army can tolerate and accept. This distinction is made to aid the system designer in allocating reliability characteristics of the device.

6.4 Design note. Cooler input power will increase when tested with a Dewar (0.4 WHL) from 50 w to 53 w at 23°C ambient temperature and from 55 w to 60 w at 71°C ambient temperature.

6.5 Subject Term (key word listings.

cooler
infrared detector,
heat sink

6.6 References.

USA Electronics Command Drawings

SM-D-969477 HD-1033B/UA	Cooler, Cryogenic Mechanical
SM-D-971500 HD-1033C/UA	Cooler, Cryogenic Mechanical
SM-D-773447	Outline Dimensions Cooler, Cryogenic Mechanical

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54490-5004372	Cryogenic Heat Load Assembly (1 watt integral)

6.7 Changes from previous issues. Marginal notations are not used in this revisions to identify changes with respect to the previous issue due to the extent of the changes.

Concluding Materials

Custodian
Army-CR

Preparing Activity
Army-CR
Project 5855-0136

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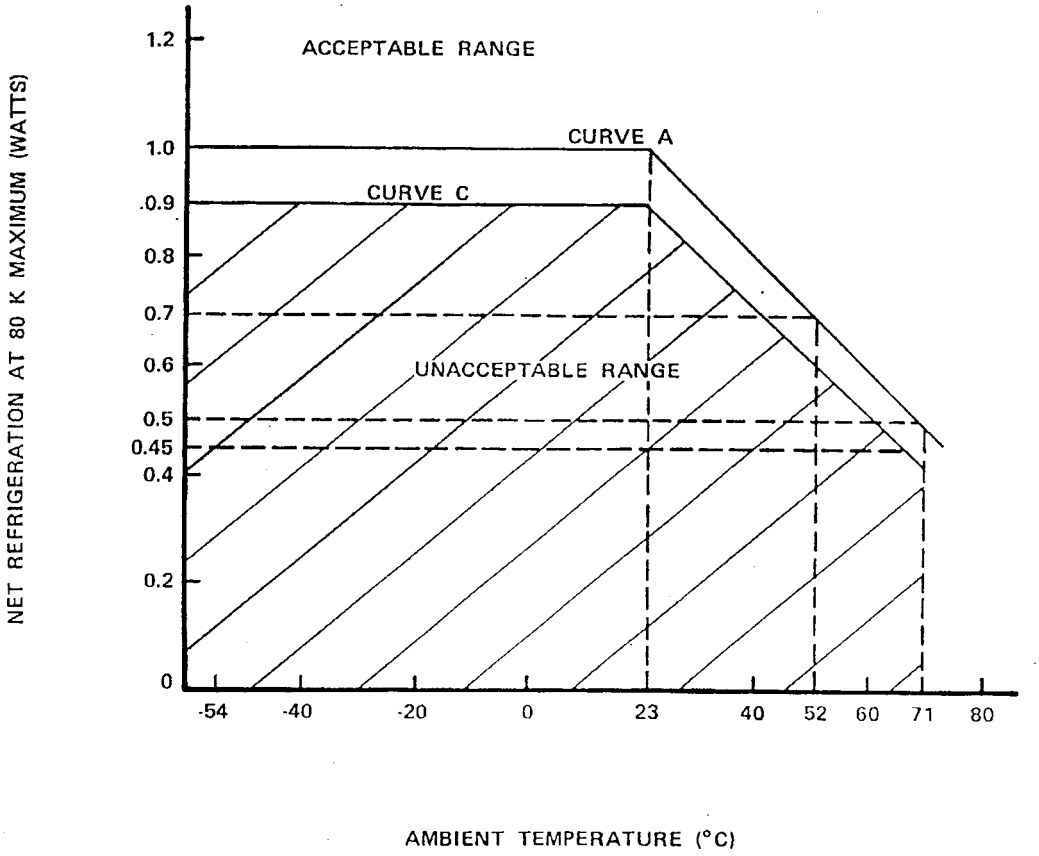


FIGURE 1. Cooling capacity.

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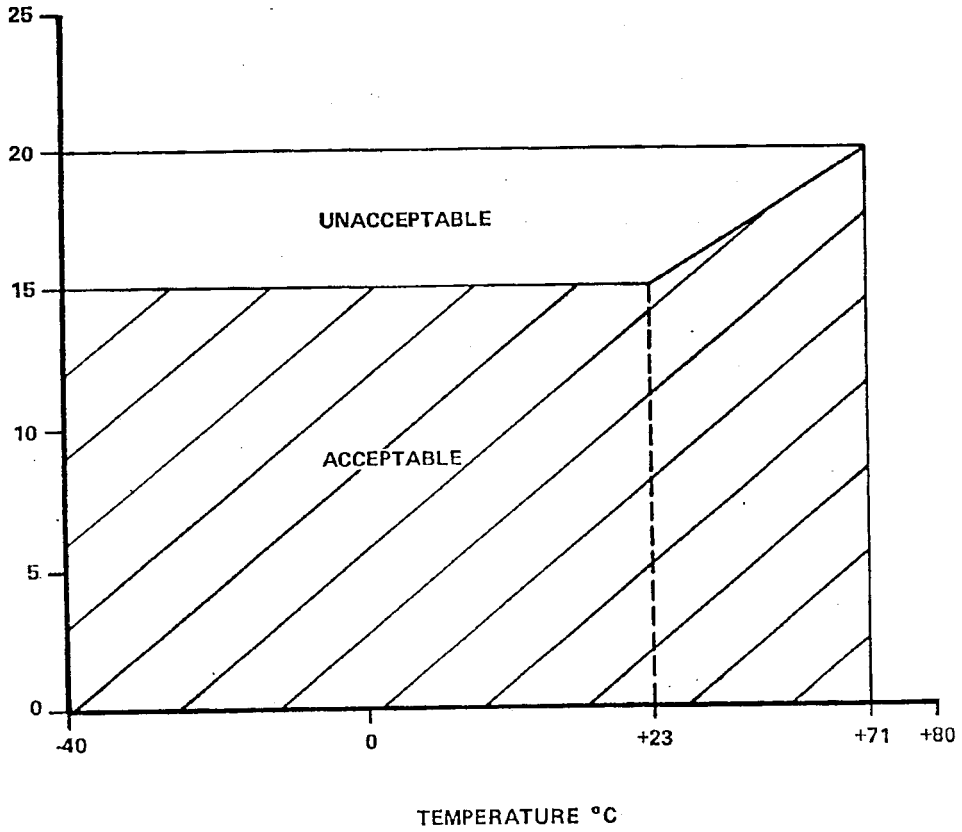


FIGURE 2. Cool down curve.

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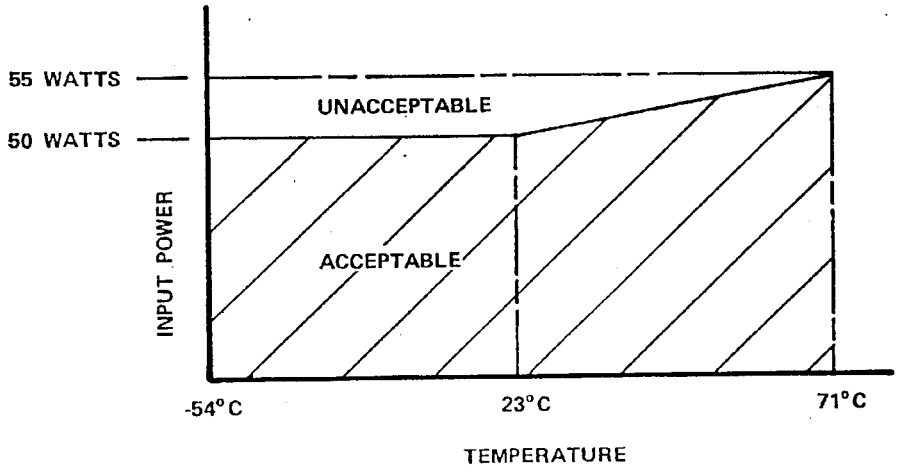


FIGURE 3. Input power with heat load applied.

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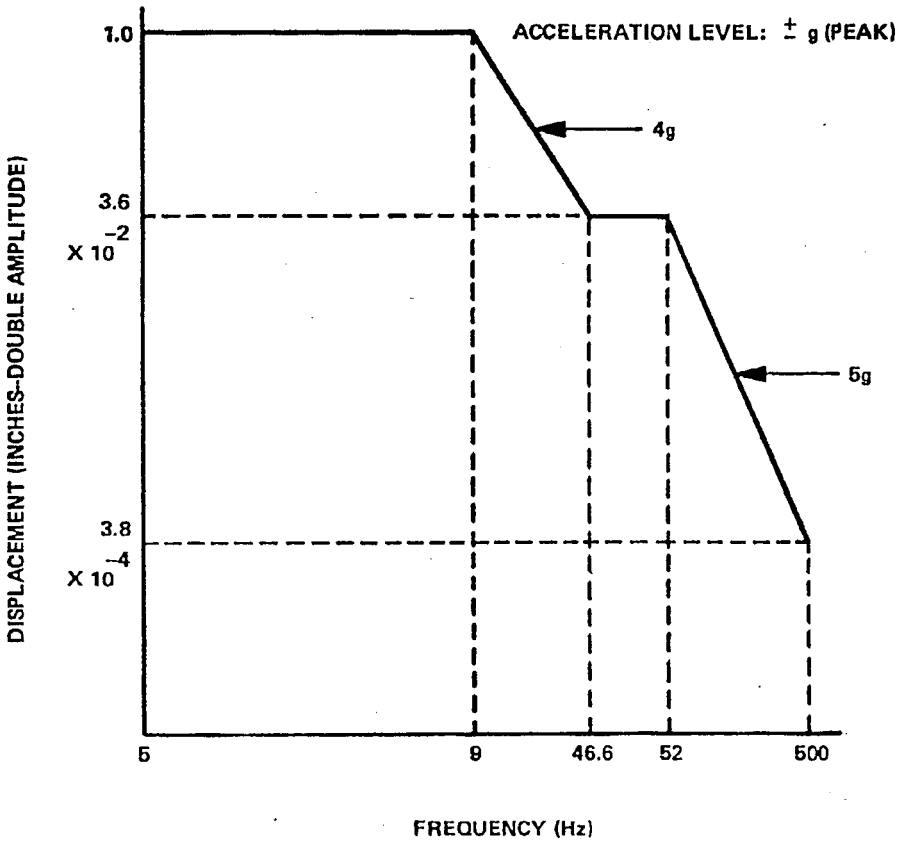
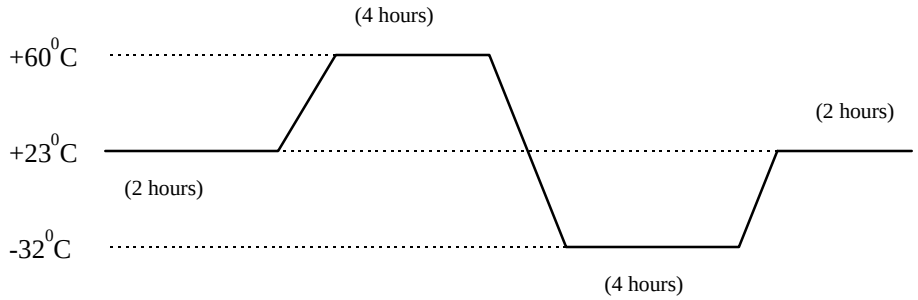


FIGURE 4. VIBRATION TEST PROFILE

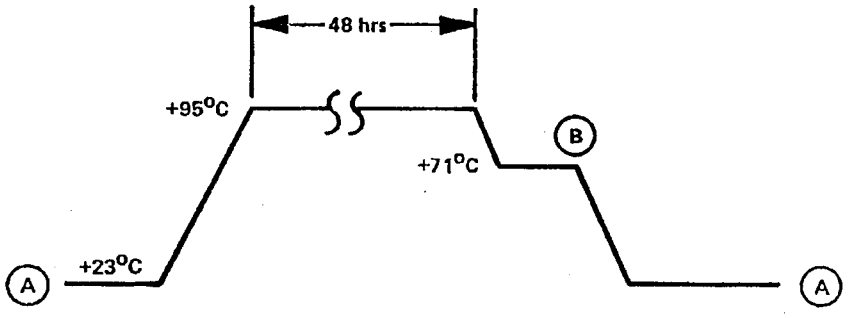
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1) Chamber temperature extremes shall not vary more than 5^{0}C per hour.
2) Rate of chamber temperature change shall average less than 5^{0}C per minute
3) Unit operates through full cycle.

FIGURE 5. Burn-in.

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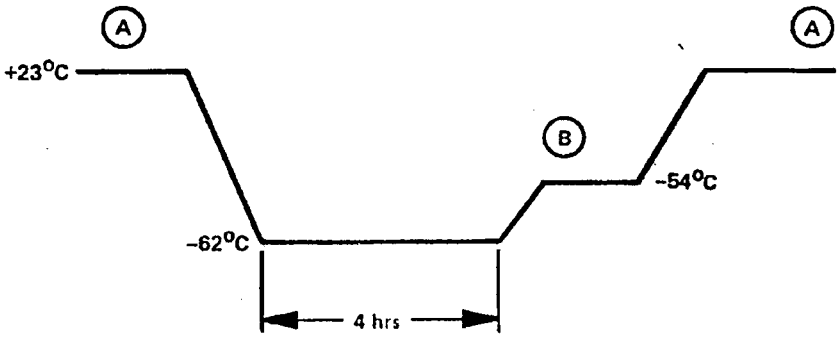


(A) GROUP A TESTS (TABLE III) ARE CONDUCTED AT THESE POINTS.

(B) COLDFINGER TEMPERATURE WITH 0.5 WATT APPLIED HEAT LOAD MONITORED AT THIS POINT TO VERIFY ABILITY OF COOLER TO MAINTAIN 80 K TEMPERATURE.

FIGURE 6. HIGH TEMPERATURE TEST PROFILE

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(A) GROUP A TESTS (TABLE III) ARE CONDUCTED AT THESE POINTS.

(B) COLDFINGER TEMPERATURE WITH 0.8 WATT APPLIED HEAT LOAD MONITORED AT THIS POINT TO VERIFY ABILITY OF COOLER TO MAINTAIN 80 K TEMPERATURE.

FIGURE 7. LOW TEMPERATURE TEST PROFILE

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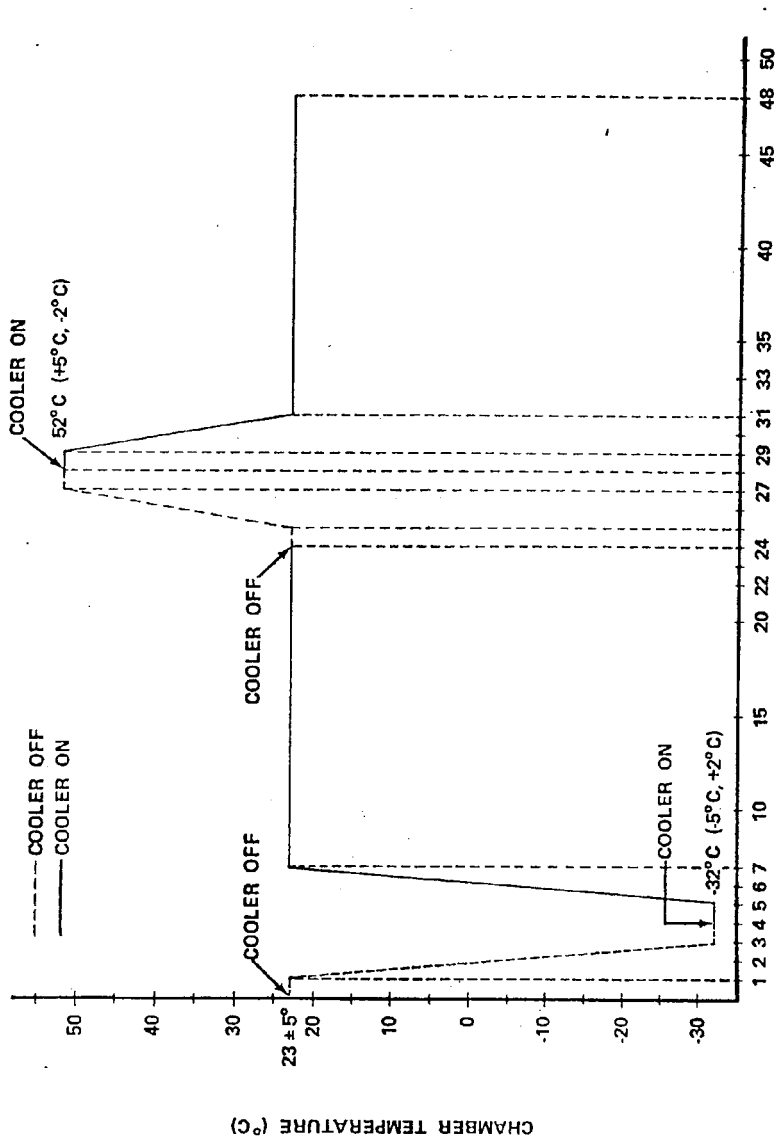


FIGURE 8. Life test.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49175C	**2. DOCUMENT DATE (YYMMDD)** **9 October** 1997

3. DOCUMENT TITLE: COOLER, CRYOGENIC, MECHANICAL HD-1033B/UA AND HD-1033C/UA

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
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c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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